Title – Food Tests Required Practical

What chemical can be used to identify if starch is present?

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DO NOW

Food Tests Required Practical

Lesson 9

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Log on to Google classroom and Complete the quiz

Due Tuesday 12th October

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HOME LEARNING

Learning Objective

- Explain the importance of cell differentiation.
- Describe how cells, tissues, organs and organ systems are organised to make up an organism.
- Understand size and scale in relation to cells, tissues, organs and body systems.

Different qualitative tests can be carried out to show the presence of different food groups.

- Iodine – No Starch – Stays Orange/Brown
- - Starch present – Blue-Black
- Benedict's Solution – No Sugars Stays Blue
- - Sugar Present – Will be a range of colour depending on sugar concentration. Green low sugar Red High sugar
- Biuret Solution – No Protein – No Change
- - Protein present – Lilac Colour Visible
- Emulsion Test (Ethanol) – No Fats – Clear
- - Fats present – will turn cloudy

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ACTIVATION

https://www.youtube.com/watch?v=akMLGbNA0gE&t=397s

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ACTIVATION

Aim – To Identify starch, reducing sugars, proteins and lipids in foods.

Complete a prediction, which foods contain which substances

Watch the demo

ACTIVATION

- For each test add 2 spatulas of each food into a test tube
- Iodine Test for Starch – Add a few drops of iodine to the food and record the colour observed
- Benedict's Test for Reducing Sugars – Add 1cm^3 of water to the food.
- Add 1cm^3 of Benedict's solution to the food.
- Place in water bath for 5 mins at 95°C then record the colour observed.
- Biuret Test for Proteins - Add 1cm^3 of water to the food.
- Add 1cm^3 of Sodium Hydroxide to the food.
- Add 2 drops of copper sulphate to the food and stir. Leave as long as possible (minimum 10 minutes) Record the colour observed
- Emulsion Test for Lipids - Add 2cm^3 of Ethanol to the food and shake vigorously. Allow to settle.
- Pour the liquid from the top of the mixture into a test tube half-filled with water. Record if the water is cloudy or clear.

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ACTIVATION

Title Food Tests Required Practical

Copy down the table into your books.

Food	Starch – Iodine - Colour	Sugar – Benedicts - Colour	Protein – Sodium Hydroxide & Copper Sulphate - Colour
Crackers			
Egg Yolk			
Egg White			
Biscuits			
Milk			
Cheese			

Food	Starch - Iodine	Sugar - Benedicts	Protein – Sodium Hydroxide & Copper Sulphate
Crackers	Black	Green	Clear
Egg Yolk	Brown	Blue	Purple
Egg White	Brown	Purple	Purple
Biscuits	Black	Orange	Clear
Milk	Brown	Yellow	Purple
Cheese	Brown	Blue	Purple

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ACTIVATION

Food	Starch - Iodine	Sugar - Benedicts	Protein – Sodium Hydroxide & Copper Sulphate	Fats/ Lipids
Crackers	Yes	Yes	No	No
Egg Yolk	No	No	Yes	Yes
Egg White	No	No	Yes	Yes
Biscuits	Yes	Yes	No	No
Milk	No	Yes	Yes	No
Cheese	No	Yes	Yes	Yes

Conclusion

1.Which food groups were found in each of the foods you tested?

2. Were any of the colour changes difficult to see? Discuss your answer.

Some observations are difficult, as the food colour may affect the colour observed; biuret change can be difficult to spot.

3. Suggest any improvements you could make to the investigation.

Improvements could include crushing the food, dissolving the food in water first, holding the tubes against a white background to see the colour change more clearly

Plenary

Identify any problems you had with this experiment.

Explain how the method could be improved to reduce or avoid these errors.